Baseband Adaptive Predistorter to Linearize Power Amplifiers With Memory

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***Abstract* — This paper presents a method intented to linearize a power amplifier with memory using a baseband adaptive predistortion technique. The power amplifier is modeled by the baseband Hammerstein time-domain model, which can take into account both the nonlinearity and the memory effect of the power amplifier. The predistorter is relatively simple and adapts to the amplifier's drifts along time. Simulation results show that the proposed method allows to eliminate totally the intermodulation products and the higher harmonic components due to amplifier distortions with a two-tone input signal. Using a modulated signal, the spectral regrowth and the constellation distortions are eliminated.**

I. INTRODUCTION

In modern mobile telecommunications systems, many different types of data can be transmitted such as speech, email, picture, video, web page, ... To transmit correctly these data and to have a good network capacity, the *UMTS* norm imposes severe constraints in term of linearity. That is why this paper aims to linearize power amplifiers.

Most of the proposed power amplifier linearization methods suppose the power amplifier to be memoryless [1]-[2], but due to the widening of the transmitted signal bandwidth, the memory effect needs to be taken into account.

Furthermore, the power amplifier characteristics vary along time due to temperature changes or ageing. Many published papers consider this effect, but they use heavy structures such as neural networks [3]-[4] or Look-Up Tables (LUT) which can use a large memory size [5]-[6].

In this paper we propose a new baseband predistortion method for linearization of a power amplifier which takes into account both the nonlinearity and the memory effect of the power amplifier. An adaptation stage is proposed to follow the drifts of the amplifier along time.

The paper is organized as follows. Section 2 is a theoretical approach of our method. Section 3 presents some simulation results and, finally, conclusions are drawn in section 4.

II. LINEARIZATION METHOD

A. Amplifier Model

The baseband Hammerstein temporal model (Fig. 1) is used to model the behavior of the power amplifier. The nonlinearity is depicted by an odd-order polynomial, which coefficients are named a_i. The memory effect of the component is modelized by a FIR filter (Finite Impulsionnal Response), its coefficients are named h_j and P is the length of its memory.

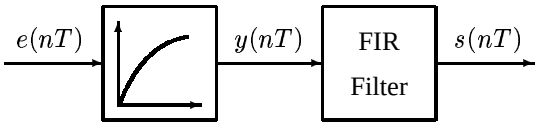


Fig. 1. Hammerstein behavioral model for the amplifier.

The output signal $s(nT)$ of the amplifier may be calculated from the input signal $e(nT)$ by (1).

$$\begin{aligned} s(nT) &= \sum_{k=0}^{P-1} \sum_{i=0}^{N} h_k a_{2i+1} |e\left[(n-k)T\right]|^{2i} e\left[(n-k)T\right] \\ &= \Gamma_{amp}\left(e(nT)\right) \end{aligned} \quad (1)$$

where Γ_{amp} represents the transfer functions of the power amplifier.

B. Predistortion theoretical Analysis

The principle of predistortion is to insert a nonlinear stage before the power amplifier in order to create a distortion inverse to that created by the power amplifier, so that globally the transfer function will be linear. The output signal can then be written in the following equation:

$$s(nT) = \Gamma_{amp}\left(\Gamma_{pred}\left[v(nT)\right]\right) = Kv(nT) \quad (2)$$

where Γ_{pred} and Γ_{amp} represent respectively the transfer functions of the predistorter and the power amplifier, and $v(nT)$ is the signal to be amplified before transmitting.

The scheme of the predistorter is shown on Fig. 2. The parameters a_i and h_j represent respectively the coefficients of the polynom describing the amplifier noninearity and the FIR filter coefficients symbolizing its memory effect.

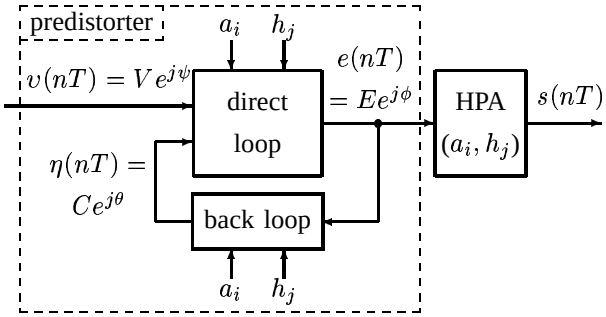


Fig. 2. General scheme of proposed predistortion technique.

From (2), it is possible to rewrite (1) by splitting static and dynamic parts. The corresponding expressions are respectively

given in (3) and (4).

$$\begin{cases} h_0 \sum_{i=0}^{N} a_{2i+1}\beta(n,i) = Kv(nT) - \eta(nT) & (3) \\ \eta((n+1)T) = \sum_{k=1}^{P-1}\sum_{i=0}^{N} h_k a_{2i+1}\beta(n+1-k,i) & (4) \end{cases}$$

where $\beta(n,i) = |e(nT)|^{2i}e(nT)$, $e(nT)$ is the signal at the output of the predistorter, K is the modulus of the amplifier gain, $v(nT)$ is the signal at the input of the predistorter and $\eta(nT)$ represents the dynamic part, which compensates the memory effect of the power amplifier. Fig. 2 remembers these notations.

Let us express the predistorted signal $e(nT)$ in its polar form: $e(nT) = E(nT)e^{j\phi(nT)}$.

Equation (3) can be written as:

$$\left[\sum_{i=0}^{N} a_{2i+1}|e(nT)|^{2i}\right] e(nT) = \frac{Kv(nT) - \eta(nT)}{h_0} \quad (5)$$

The calculation of the modulus of this equation leads to:

$$\left|\sum_{i=0}^{N} a_{2i+1}E^{2i+1}\right|^2 = \frac{|Kv(nT) - \eta(nT)|^2}{|h_0|^2} \quad (6)$$

The modulus of the predistorted signal $E(nT)$ is any one of the real and positive roots of the polynom of (6). The phasis of this signal is given by (7), obtained from (5).

$$\phi(nT) = Arg\left\{\frac{Kv(nT) - \eta(nT)}{h_0\left[\sum_{\text{k odd}} a_k E^k(nT)\right]}\right\} \quad (7)$$

Finally, the equations of the predistorter are:

$$\begin{cases} e(nT) = E(nT)e^{j\phi(nT)} & (8) \\ \eta((n+1)T) = \sum_{k=1}^{P-1}\sum_{i=0}^{N} h_k a_{2i+1}\beta(n+1-k,i) & (9) \end{cases}$$

where $\beta(n,i) = |e(nT)|^{2i}e(nT)$, $E(nT)$ and $\phi(nT)$ are calculated respectively from (6) and (7).

C. Adaptation Technique

The proposed predistorter needs the values of the coefficients of the amplifier model ([7]-[8]), as shown in Fig. 2. In order to make the system autonomous and to follow the eventual drifts of the amplifier due to temperature or ageing, these coefficients of nonlinearity (a_i) and memory effect (h_j) must be updated regularly. If Γ_{amp} is the amplifier transfer function, then the adaptation consists to minimize the term:

$$J(nT) = \sum_{i=1}^{n} |s(iT) - \Gamma_{amp}(e(iT))|^2 \quad (10)$$

The identification step is realized as follows. (1) can be written in a form of matrix $AX = B$ where:

$$A^T = \begin{pmatrix} e(0) & \cdots & e(n) \\ |e(0)|^2 e(0) & \cdots & |e(n)|^2 e(n) \\ \vdots & \cdots & \vdots \\ |e(0)|^{2N} e(0) & \cdots & |e(n)|^{2N} e(n) \\ 0 & \cdots & e(n-1) \\ \vdots & \cdots & \vdots \\ 0 & \cdots & |e(n-P)|^{2N} e(n-P) \end{pmatrix}$$

$$X = \begin{pmatrix} h_0 a_1 \\ h_0 a_3 \\ \vdots \\ h_0 a_{2N+1} \\ h_1 a_1 \\ \vdots \\ h_{P-1} a_{2N+1} \end{pmatrix} \qquad B = \begin{pmatrix} s(0) \\ s(1) \\ \vdots \\ s(n) \end{pmatrix}$$

We remember that $e(nT)$ and $s(nT)$ are respectively the signals at the input and the output of the power amplifier, without predistorter, at instant n.

As the number of samples is high, this system is overdetermined and a classical least square resolution can be used to find the solutions:

$$X = \left(A^H A\right)^{-1} A^H B \quad (11)$$

Once the solution for X is obtained, we impose $h_0 = 1$. Thus, the N first elements of X give directly the nonlinearity coefficients $\tilde{a}_i$. The other elements of X give $(N+1)$ solutions for each memory effect coefficient $\tilde{h}_j$. As we have a system which can be written under the form $\tilde{A}\tilde{h}_j = X_j$, where X_j contains the elements of X corresponding to $\tilde{h}_j$ and $\tilde{A}$ contains the $(N+1)$ nonlinearity coefficients, it is possible to determine the optimum value of $\tilde{h}_j$ by solving equation (12).

$$\tilde{h}_j = \left(\tilde{A}^H \tilde{A}\right)^{-1} \tilde{A}^H X_j \quad , j = 0 \ldots P-1 \quad (12)$$

where:

$$X_j = \begin{pmatrix} \tilde{h}_j \tilde{a}_1 \\ \tilde{h}_j \tilde{a}_3 \\ \vdots \\ \tilde{h}_j \tilde{a}_{2N+1} \end{pmatrix} \quad and \quad \tilde{A} = \begin{pmatrix} \tilde{a}_1 \\ \tilde{a}_3 \\ \vdots \\ \tilde{a}_{2N+1} \end{pmatrix}$$

These identificated coefficients are then used in the predistorter to linearize the power amplifier. The global adaptive linearisation scheme is presented in Fig. 3.

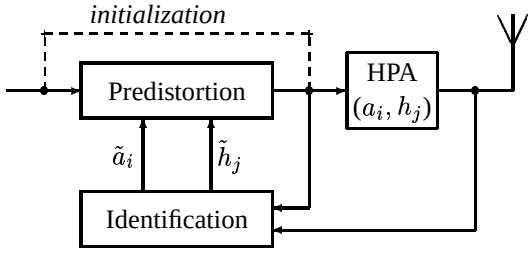


Fig. 3. Global adaptive linearisation scheme.

III. Simulation Results

In the following simulations, we used the initial coefficients presented in Table I for the Hammerstein model of the power amplifier.

Coefficient	real part	imaginary part
a_1	1.96	-0.88
a_3	$3.91.10^{-8}$	$2.58.10^{-8}$
a_5	$-2.92.10^{-14}$	$-8.16.10^{-15}$
h_0	1	0
h_1	0.052	0.016
h_2	-0.013	-0.020
h_3	0.017	0.006
h_4	-0.005	0.001

TABLE I

Initial coefficients of the amplifier model

The AM-AM and gain compression characteristics for different frequencies corresponding to this amplifier are respectively presented in Fig. 4 and 5. As shown on these figures, the frequency spread is relatively weak. The gain of the amplifier is around $7dB$ and the $1dB$ compression point corresponds to an input power of $18.8dBm$.

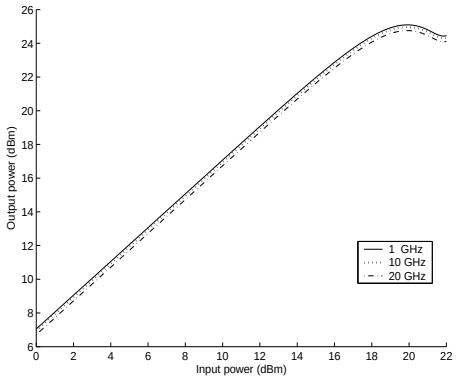


Fig. 4. AM-AM characteristic of the simulated amplifier

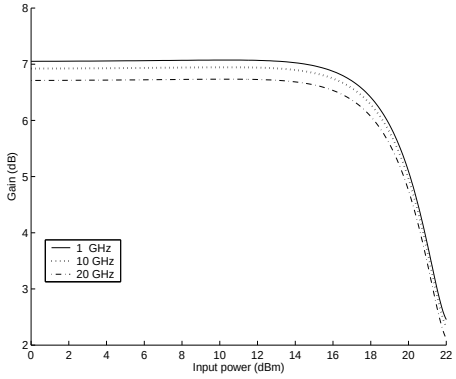


Fig. 5. Gain compression characteristic of the simulated amplifier

As the predistortion step needs the coeficients of the model, it is necessary to evaluate them before linearizing the amplifier. That is why a part of the input signal is applied directly to the amplifier (without predistorter), as shown on Fig. 3. Once the coefficients are estimated, it is possible to linearize the amplifier and to update them.

A. Two-tone Signal

In the first simulation, we applied 50 frames of 8700 samples of a two-tone signal at the input of the system. The carrier frequencies are $f_1 = 1.45GHz$ and $f_2 = 1.5GHz$. The input signal power is $Pe = 15dBm$ and each coefficient of the model can vary from one frame to the other up to 1% from its previous value following a uniform law. Results are presented in Fig.6. In the case where the amplifier is used alone (without linearization), intermodulation products and higher harmonic appear. Using the proposed predistorter, these parasite components are totally eliminated and the noise level is largely reduced, even if the adaptation stage is not used.

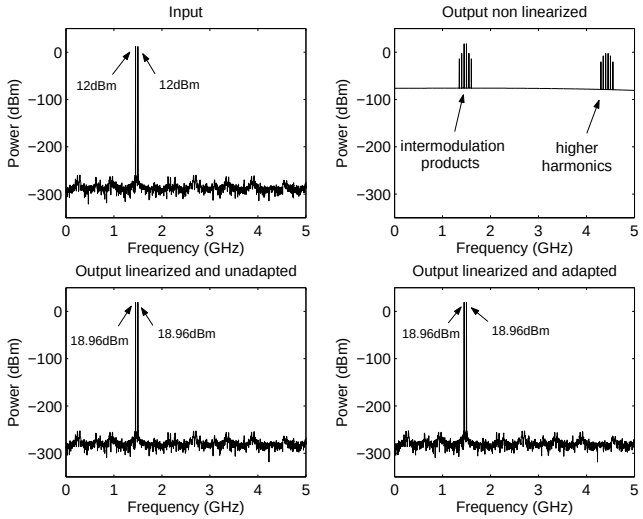


Fig. 6. Results with a two-tone signal

The Fig. 7 presents the relative errors on the coefficients due to identification process. These relative errors are between $10^{-9}\%$ and $10^{-8}\%$ for the nonlinearity coefficients and around $10^{-7}\%$ for the memory effect ones.

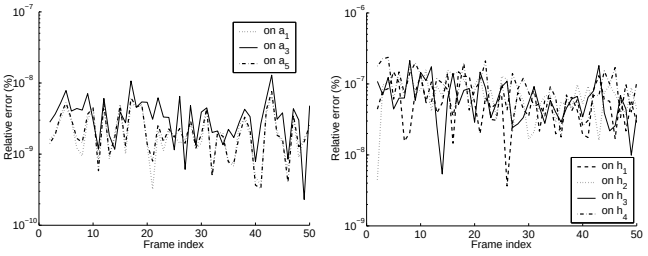


Fig. 7. Relative identification errors for nonlinearity coefficients (left) and memory effect coefficients (right) with a two-tone input signal.

B. 64-QAM Signal

The same simulation is realized using 50 frames of 25600 samples of a 64-QAM signal. The input power is $Pe = 15dBm$. As for two-tone case, we compare the case without linearization to the results obtained using the predistorter, and the coefficients of the amplifier model vary on the same law as previously. Fig. 8 and 9 respectively present the spectra and the constellation diagrams with and without linearization.

On Fig. 8 we can observe spectral regrowth due to the amplifier distortions in the case where the predistorter is not

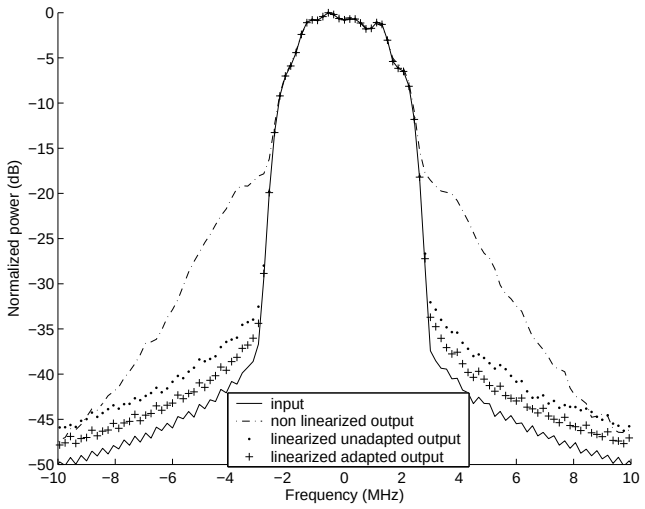


Fig. 8. Spectra for a 64-QAM input signal

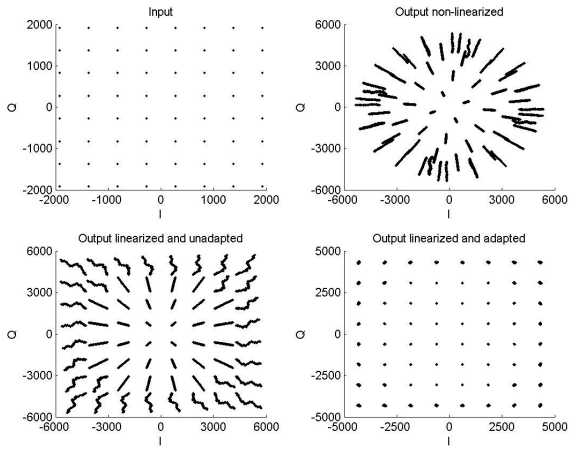


Fig. 9. Constellation diagrams for a 64-QAM input signal

used, what leads to the rising of ACPR and adjacent channel interference. If the proposed linearization technique is used this spectral regrowth is greatly reduced, particularly when the coefficients are regularly updated.

Fig. 9 shows that the constellation diagram is completely distorted without predistorter, which will generate transmission errors. With the proposed predistorter, the constellation is aligned but if identification is not used, the coefficient variations affect the diagram. Updating regularly the coefficients, the constellation diagram is well aligned and bit error rate (BER) is minimized.

The relative errors on the coefficients due to identification process are presented in Fig. 10. These relative errors are between $10^{-8}\%$ and $10^{-6}\%$ for the nonlinearity coefficients and around $10^{-6}\%$ for the memory effect ones.

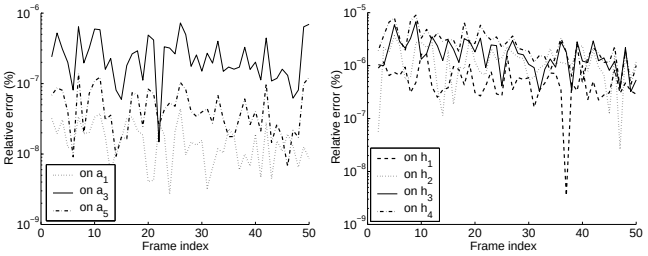


Fig. 10. Relative identification errors for nonlinearity coefficients (left) and memory effect coefficients (right) with a 64QAM input signal.

As in the case of the two-tone signal these identification errors are very small and, as seen in Fig. 8 and 9, sufficiently small to linearize correctly the power amplifier.

IV. Conclusion

The proposed linearization method, based on the baseband temporal Hammerstein model, allows to eliminate intermodulation products, higher harmonics and spectral regrowth due to power amplifier nonlinearities. The constellation diagram is also realigned, what reduces transmission errors and minimizes bit error rate. Moreover the system is totally adaptive and the necessity of an adaptation function has been demonstrated. A measuring bench is actually under development in my laboratory to test this linearization method by adaptive predistortion with a real power amplifier.

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